

Design of a multifunctional integrated optical isolator switch based on nonlinear and nonreciprocal effects

Licinius D. S. Alcantara

Marcos A. C. Lima

Amílcar C. César

Ben-Hur V. Borges

University of São Paulo

São Carlos School of Engineering

Electrical Engineering Department

Av. Trabalhador São-Carlense, 400

13566-590 São Carlos, SP, Brazil

E-mail: benhur@sel.eesc.usp.br

Fernando L. Teixeira

The Ohio State University

Department of Electrical and

Computer Engineering

ElectroScience Laboratory

1320 Kinnear Road

Columbus, Ohio 43212

Abstract. The design of a new multifunctional integrated optical device capable of operating simultaneously as an optical switch and optical isolator is presented. The device consists of a multilayer directional coupler employing one nonlinear and one magneto-optic layer, with all other layers being linear and isotropic. Therefore, only phase matching conditions are required for the device operation irrespective of the propagation directions and operation regime (linear/nonlinear). The structure is optimized via a genetic algorithm and tested with a numerical formalism based on the wide-angle finite-difference beam propagation method. The results clearly demonstrate the ability of performing two distinct functions on the same device. © 2005 Society of Photo-Optical Instrumentation Engineers. [DOI: 10.1117/1.2139655]

Subject terms: nonlinear medium; magneto-optic medium; beam propagation method; finite difference; integrated optics; optimization; genetic algorithm.

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1 Introduction

Optical isolators and switches are crucial components in optical communication systems, employed fundamentally as a protection against undesired reflections back into the source, and as signal routers, respectively. So far, these devices can only be found separately. Thin film optical isolators normally require a precise control of its geometrical parameters, and this requirement has been one of the major causes for the relatively smaller effort in investigating this configuration. A comprehensive review of the current status of magneto-optic isolators can be found in Ref. 1. Most integrated optics isolators currently available are based on the nonreciprocal TE-TM mode conversion, which requires a precise phase matching condition between both modes (form birefringence).^{2,3} A quite interesting isolator configuration was proposed by Zhuromskyy et al.^{4,5} The device in this case was based on a Mach-Zehnder configuration with one branch operating as a nonreciprocal phase-shifter for TE modes and the other as a nonreciprocal phase-shifter for TM modes. Even though this structure is less sensitive to geometrical variations, it still suffers from the different loss magnitudes for TE and TM modes. A similar approach was followed by Fujita et al.⁶ More recently, Lohmeyer et al. have introduced the cross strip magneto-optic isolator.⁷ Differently from previous configurations, this structure does not require the use of additional components such as polarizers or power splitters.

Nonlinear waveguide devices, on the other hand, have been extensively investigated in the literature in the last years.⁸⁻¹¹ The interest for optical switching in particular has been driven by the continuously growing traffic of optical

networks. Growing demand for traffic capacity exposes a serious limitation of electronic based switching and highlights the need for all-optical switching, where optical-electronic and electro-optical conversions are no longer necessary.

In many cases, the insertion loss caused by the inclusion of an optical isolator or switch in an optical communication link may be too severe. In this scenario, the design of multifunctional optical devices capable of performing multiple tasks simultaneously will allow a considerable increase in efficiency, with a consequent reduction of costs and maintenance in future optical communication systems. The fabrication of optical devices made of nonlinear and nonreciprocal materials simultaneously paves the way for obtaining such functions in a single device.

The idea of combining nonlinear and nonreciprocal materials was originally proposed by Gnideck¹² to obtain an optical isolator and switch on the same device. This analysis was based on a dispersion relation obtained for a three-layer planar waveguide. This allowed for the investigation of the isolation and switching characteristics as a function of the input optical power density. This concept was further investigated by Hutchings,¹³ for different waveguide structures. The polarization evolution was computed via a Hamiltonian formalism employing the Stokes parameters. More recently, the analysis of such media combination has been carried out by a finite-difference (FD) based beam propagation method (BPM) by Alcantara et al.¹⁴ In this case, the FD-BPM formalism was extended to account for any nonlinear mechanisms.

This paper presents the design of a new multifunctional integrated optical device capable of performing simultaneous optical isolation and optical switching. This device operates at TM polarization and is basically a directional

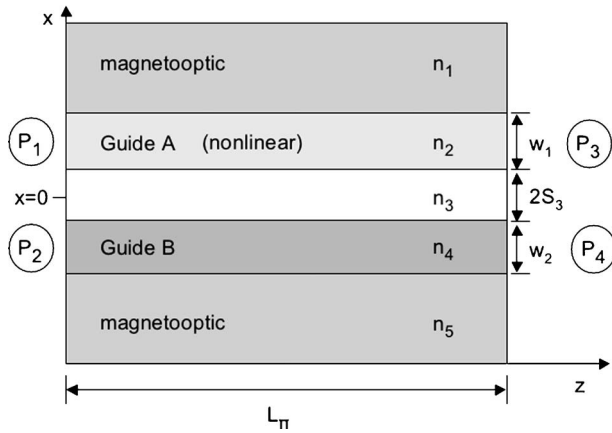


Fig. 1 Longitudinal view of the proposed nonlinear-nonreciprocal structure. $P_{1,2,3,4}$ designates the ports of the device, L_π is the coupling length. Layers n_3 and n_4 are isotropic and linear.

coupler. The energy transfer between the waveguides is controlled by the optical power density launched at the input port of the device via nonlinear effect. The field excitation at the input port of the coupler is assumed as continuous wave (cw). The hybrid characteristic of the structure is provided by the nonreciprocal effect, which prevents any reflection from reaching the optical source at the input port. The analysis and design of this device is carried out via the wide-angle FD-BPM, described in more detail in Alcantara et al.,¹⁴ and a genetic algorithm (GA). The present FD-BPM formalism has the unique advantage of taking into account any nonlinearity mechanism as well as longitudinal variations in the refractive index. The GA is utilized to optimize the geometry and performance of the nonlinear/nonreciprocal directional coupler, for a fixed set of material parameters.

2 Proposed Structure

The longitudinal view of the proposed structure is shown in Fig. 1(a), where $P_{1,2,3,4}$ designates the ports of the device and L_π the coupling length. The physical and geometrical parameters of the device are summarized in Table 1. The parameters δ , γ , and n_{2s} will be properly introduced in Sec. 3. In this structure, the magneto-optic layers n_1 and n_5 act as

Table 1 Physical and geometrical parameters for the structure depicted in Fig. 1. The wavelength $\lambda = 1.55 \mu\text{m}$.

Layer	Refractive index	Thickness (μm)	δ	n_{2s} (m^2/W)
n_1	2.23	7.0	-0.019	0
n_2	2.26	1.2	0	10^{-9}
n_3	2.00	To be optimized	0	0
n_4	2.26	1.23	0	0
n_5	2.23	7.0	-0.019	0

Power level	Direction of wave propagation	
	+z	-z
Low (P_1)		
High (P_2)		

Fig. 2 Requirements for the nonlinear and nonreciprocal coupler design.

cladding layers, layers n_2 (nonlinear) and n_4 (linear) act as guiding layers, and layer n_3 (linear) acts as a separation layer.

The intended behavior of this device for the forward and backward propagation directions is depicted in Fig. 2 for two distinct optical power levels. The operation of the coupler can be summarized as follows, assuming that light is launched into port 1. Under low optical power density, P_1 , the signal coming from port 1 is coupled to guide B and emerges on port 4. If, by any chance, light is reflected back from port 4, it will be deviated to port 2 via nonreciprocal effect. This procedure prevents the optical source in port 1 from suffering any undesirable effect due to reflected power, such as frequency chirping. Under high optical power density, P_2 , guides A and B are phase mismatched and light is propagated from port 1 to port 3. If light is reflected from port 3, the phase matching condition for guides A and B will be enforced by the nonreciprocal effect and light will be deviated to port 2, keeping port 1 isolated. Therefore, this device works as an optical switching commuting light from port 1 to port 4 (for low power levels), or to port 3 (for high power levels). Simultaneously, port 1 is always kept isolated from any reflections via nonreciprocal effect.

3 Material Parameters

The dielectric permittivity tensor for a magneto-optic medium submitted to a dc magnetostatic field applied along the y axis is given by:

$$\bar{\epsilon}^{nr} = \epsilon_0 \begin{bmatrix} n_x^2 & 0 & j\delta \\ 0 & n_y^2 & 0 \\ -j\delta & 0 & n_z^2 \end{bmatrix}, \quad (1)$$

where n_x , n_y , and n_z are the refractive indices in the x , y , and z directions, respectively, δ is the first-order magneto-optic effect, and ϵ_0 is the vacuum dielectric permittivity. The permittivity tensor in the nonlinear medium depends on the electric field intensity as follows,

$$\epsilon_{xx}^{nl} = \epsilon_c^{nl} + \alpha(|E_x|^2 + \gamma|E_z|^2), \quad (2)$$

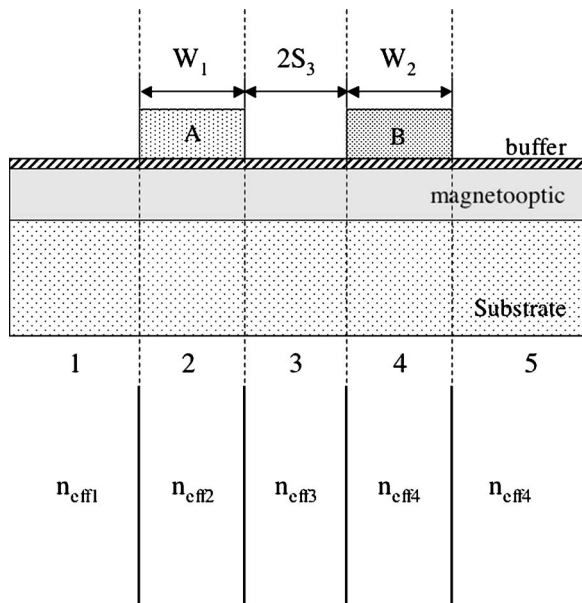


Fig. 3 Reduction of a three-dimensional structure to a two-dimensional one. The new refractive indices ($n_{\text{eff}} = \beta/k_0$) are assumed to be invariant along the z axis. The layers A and B are considered as nonlinear and linear media, respectively.

$$\varepsilon_{zz}^{nl} = \varepsilon_c^{nl} + \alpha(\gamma|E_x|^2 + |E_z|^2), \quad (3)$$

where γ is a coefficient that depends on the nonlinearity mechanism⁵ ($\gamma=1$ for thermal effects or electrostriction, $\gamma=1/3$ for electronic distortion, and $\gamma=-1/2$ for molecular orientation). The ε_c^{nl} parameter is the nonlinear material static permittivity, α is the nonlinear coefficient given by $\alpha = \varepsilon_c^{nl} n_{2s} c \varepsilon_0$, where c is the light velocity in vacuum and the constant $n_{2s} = 10^{-9} \text{ m}^2/\text{W}$ (focusing media).⁶

In magneto-optic media, the anisotropy is introduced by the presence of a magnetostatic field (H_{DC}) that, when conveniently oriented, allows one to separate the solutions of Maxwell equation in TE and TM modes.

It should be pointed out that the fabrication of the proposed structure may pose some difficulties for the choice of materials at this present moment, particularly for the nonlinear medium. Depending on the material combination, a vertical structure such as in Fig. 1 may not be best choice due to growth difficulties. Fortunately, it is well known that any tri-dimensional optical waveguide structure can be reduced to a two-dimensional one by means of an effective index approach. This approach can be visualized in Fig. 3 for a hypothetical configuration of the coupler. The buffer layer is considered here for the case when the crystalline structure of the top layers needs to be maintained, but it should be made as thin as possible. This structure is divided in five different regions, where each region is treated as an isolated planar waveguide, resulting in five effective indices, $n_{\text{eff}} = \beta/k_0$ (where β is the longitudinal propagation constant and $k_0 = 2\pi/\lambda_0$). These five indices will now define the index profile of the new two-dimensional structure whose layer thicknesses are assumed as the width of each individual region. This new structure configuration falls exactly in the case of Fig. 1, and can be readily investigated with the approach adopted in this work. Additionally, the

Table 2 Inverse permittivity elements.

Parameter	Isotropic	Magneto-optic	Nonlinear
ξ_{xx}	$\frac{1}{\varepsilon_0 n^2}$	$\frac{1}{\varepsilon_0 n_x^2 n_z^2 - \delta^2}$	$\frac{1}{\varepsilon_0 \varepsilon_{xx}^{nl}}$
ξ_{zz}	$\frac{1}{\varepsilon_0 n^2}$	$\frac{1}{\varepsilon_0 n_x^2 n_z^2 - \delta^2}$	$\frac{1}{\varepsilon_0 \varepsilon_{zz}^{nl}}$
ξ_{xz}	0	$-\frac{1}{\varepsilon_0 n_x^2 n_z^2 - \delta^2} \frac{j\delta}{\delta}$	0

formalism introduced in Sec. 5 is sufficiently general to account for any combination of magneto-optic and linear media, i.e., the location of these layers may be chosen in such a way as to satisfy any given fabrication constraint. This aspect is important when optimization of complex waveguiding structures is required. The refractive indices and thicknesses adopted for the structure simulated in this work are intended to provide a proof of concept for the combined nonreciprocal and nonlinear effects.

4 FD-BPM Formalism for Nonreciprocal and Nonlinear Media

The FD-BPM formalism is summarized in this section. A more detailed description can be found in Ref. 14. The wave equation for the H_y component of the TM mode (H_y , E_x , and E_z) propagating along the z direction with $\partial/\partial y \equiv 0$ can be obtained from Maxwell's equations, and is written as

$$\mu_0 \frac{\partial^2 H_y}{\partial t^2} = \xi_{xx} \frac{\partial^2 H_y}{\partial z^2} + \left(\frac{\partial \xi_{xx}}{\partial z} + \frac{\partial \xi_{xz}}{\partial x} \right) \frac{\partial H_y}{\partial z} + \xi_{zz} \frac{\partial^2 H_y}{\partial x^2} + \left(\frac{\partial \xi_{zz}}{\partial x} - \frac{\partial \xi_{xz}}{\partial z} \right) \frac{\partial H_y}{\partial x}, \quad (4)$$

where μ_0 is the vacuum magnetic permeability and $\xi_{u,v}$ ($u, v = x, z$) are the elements of the inverse permittivity matrix in each medium as listed in Table 2. A static magnetic field is present along the y direction to produce magneto-optic effects. Equation (4) can be successfully applied to any combination of isotropic, magneto-optic, or nonlinear materials.

The H_y component can be written as $H_y(x, z, t) = \Psi(x, z) \exp[j(\omega t - n_0 k_0 z)]$, where ω is the signal angular frequency, k_0 is the propagation constant in free space, and n_0 is a reference effective index. Therefore, Eq. (4) becomes

$$\frac{\partial^2 \Psi}{\partial z^2} - 2j\zeta \frac{\partial \Psi}{\partial z} + P\Psi = 0, \quad (5)$$

where

$$\zeta = \kappa + \frac{j}{2} \left(\frac{\partial \ln \xi_{xx}}{\partial z} + \frac{1}{\xi_{xx}} \frac{\partial \xi_{xz}}{\partial x} \right),$$

$$P = \frac{\xi_{zz}}{\xi_{xx}} \frac{\partial^2}{\partial x^2} + \frac{1}{\xi_{xx}} \left(\frac{\partial \xi_{xx}}{\partial x} - \frac{\partial \xi_{xz}}{\partial z} \right) \frac{\partial}{\partial x} - j\kappa \left(\frac{\partial \ln \xi_{xx}}{\partial z} + \frac{1}{\xi_{xx}} \frac{\partial \xi_{xz}}{\partial x} \right) + \frac{\omega^2 \mu_0}{\xi_{xx}} - \kappa^2,$$

with $\kappa = k_0 n_0$. Equation (5) can be decomposed into two independent equations, one for each propagation direction. For the $+z$ direction, one obtains

$$\frac{\partial \Psi}{\partial z} = -j(\sqrt{P + \zeta^2} - \zeta)\Psi. \quad (6)$$

Unfortunately, the presence of the square root term on the right-hand side (r.h.s.) of Eq. (6) poses an obstacle for the finite-difference expansion. A possible way of circumventing this problem consists in expanding the r.h.s. in terms of Padé approximants of order (1,1). Therefore,

$$\sqrt{P + \zeta^2} - \zeta \approx \frac{P}{2\zeta} \left/ \left(1 + \frac{P}{4\zeta^2} \right) \right. \quad (7)$$

Higher order approximants can also be used at the expense of significantly more complicated expressions. Substituting Eq. (7) into Eq. (6) results in an equation that can be discretized by a finite-difference approach based on the Crank-Nicolson (CN) scheme. An iterative procedure is adopted in the formalism in order to accelerate convergence of the electric field and the dielectric permittivity at $z = (m + 1)\delta z$. This is necessary since the expressions for the permittivity tensor in the nonlinear region depend on the field intensity according to Eqs. (2) and (3). The boundary condition applied on the edge of the computational window is the transparent boundary condition (TBC), described in more detail in Ref. 15.

The modal effective index (β/k_0) can be directly calculated from the field profile obtained with the FD-BPM at any position along the longitudinal direction. To do so, all it needs to be done is to substitute the field in the following characteristic equation for β ,

$$\left(\int \xi_{xx} |H_y|^2 dx \right) \beta^2 + \left[j \int \left(\frac{\partial \xi_{xx}}{\partial z} + \frac{\partial \xi_{xz}}{\partial x} \right) |H_y|^2 dx \right] \beta - \left[\int \xi_{zz} \frac{\partial^2 H_y}{\partial x^2} H_y^* dx + \int \left(\frac{\partial \xi_{zz}}{\partial x} + \frac{\partial \xi_{xz}}{\partial z} \right) \frac{\partial H_y}{\partial x} H_y^* dx + \omega^2 \mu_0 \int |H_y|^2 dx \right] = 0.$$

Observe that this equation is nothing but a second-order equation for the unknown β , which can also be written as

$a\beta^2 + b\beta - c = 0$, where a , b , and c are just constants. Only the positive root for β has a physical meaning.

5 Nonreciprocal Multilayer Formalism

Optimization algorithms such as GA are more efficiently applied when a fitness function is obtained analytically. Therefore, this section presents an analytical formalism for the calculation of the forward and backward propagation constants, which are utilized for obtaining the corresponding coupling lengths. In order to make the formalism as general as possible, the multilayer structure in Fig. 1 is considered as a five-layer nonreciprocal coupler in this section, with all layers assumed magneto-optic. This allows the designer to choose freely which of the five layers will actually be magneto-optic. In that case, the magneto-optical parameters of the remaining layers should be set to zero. Nonlinear media are not included at this stage. The nonlinearity will be properly considered in the coupler design via the FD-BPM formalism as described in Sec. 7.

From Eq. (4) and assuming a time harmonic dependence $\exp(j\omega t)$, one obtains the following wave equation for nonreciprocal media in terms of the H_y component

$$\frac{\partial^2 H_y}{\partial x^2} + \left(\frac{k_0^2 - \xi_{xx}\beta^2}{\xi_{zz}} \right) H_y = 0, \quad (8)$$

where

$$\xi_{xx} = \frac{\epsilon_{zz}}{\epsilon_{xx}\epsilon_{zz} - \delta^2}, \quad \xi_{zz} = \frac{\epsilon_{xx}}{\epsilon_{xx}\epsilon_{zz} - \delta^2}.$$

Solving Eq. (8) in each layer and making use of the proper radiation conditions, one obtains

$$H_y = A_1 \exp[-\gamma_1(x - S_3 - d_2)], \quad x \geq S_3 + d_2$$

$$H_y = A_2 \cos[\kappa_2(x - S_3 - d_2/2)] + A_3 \sin[\kappa_2(x - S_3 - d_2/2)], \quad S_3 + d_2 \geq x \geq S_3$$

$$H_y = A_4 \exp[-\gamma_3 x] + A_5 \exp[\gamma_3 x], \quad -S_3 \leq x \leq S_3$$

$$H_y = A_6 \cos[\kappa_4(x + S_3 + d_4/2)] + A_7 \sin[\kappa_4(x + S_3 + d_4/2)], \quad -S_3 - d_4 \leq x \leq -S_3$$

$$H_y = A_8 \exp[\gamma_5(x + S_3 + d_4)], \quad x \leq -S_3 - d_4,$$

where A_1 through A_8 are constants to be determined, κ_i and γ_j are given by

$$\kappa_i = \sqrt{\frac{k_0^2 - \xi_{xx}^{(i)}\beta^2}{\xi_{zz}^{(i)}}}, \quad i = 2, 4$$

$$\gamma_j = \sqrt{\frac{\xi_{xx}^{(j)}\beta^2 - k_0^2}{\xi_{zz}^{(j)}}}, \quad j = 1, 3, 5$$

k_0 is the vacuum wavenumber, $\xi_{xx} = \epsilon_{zz}/\Phi$, and $\xi_{zz} = \epsilon_{xx}/\Phi$, with $\Phi = \epsilon_{xx}\epsilon_{zz} - \delta^2$. The electric field components E_x and E_z can be directly obtained with

β_a^+	β_b^+	β_a^-	β_b^-	$2S_3$
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Fig. 4 Representation of an individual formed by various genes. Each gene represents a value corresponding to the propagation constant and/or to the separation thickness. β_a and β_b are the propagation constants for the symmetric and antisymmetric modes for the multilayer structure in Fig. 1. The signs (+) and (−) indicate the propagation direction.

$$E_x = \frac{1}{\epsilon_0} \left(\xi_{xx} \frac{\beta}{\omega} H_y + j \frac{\xi_{zx}}{\omega} \frac{\partial H_y}{\partial x} \right)$$

and

$$E_z = \frac{1}{\epsilon_0} \left(\xi_{zx} \frac{\beta}{\omega} H_y - j \frac{\xi_{zz}}{\omega} \frac{\partial H_y}{\partial x} \right),$$

with $\xi_{zx} = j\delta/\Phi$. Applying the boundary conditions for the tangential components H_y and E_z , one obtains a system of eight equations and eight unknowns, which can be conveniently written in matrix form as follows

$$[M(\beta)][A] = 0. \quad (9)$$

Here, $[M(\beta)]$ is 8×8 matrix that depends only on the unknown longitudinal propagation constant β . The nonzero elements of this matrix are listed in the appendix. The propagation constant β can be easily found by solving the $\text{Det}[M(\beta)] = 0$.

6 Genetic Algorithm

The GA employed in this work utilizes simulated binary crossover (SBX) operator without any mutation operator.¹⁶ In the present case, the propagation constants and the guide separation $2S_3$ are represented as genes. Each gene is represented by a real valued system variable. The group of all genes that composes an individual is illustrated in Fig. 4. The objective function is related to the fitness function, and its value reflects how adjusted is an individual with respect to the solution of the problem. In a GA-based optimization, one can obtain several different solutions. Therefore, it is necessary to find the best solution in an acceptable time interval. The total fitness function is calculated evaluating the $\text{Det}[M(\beta)] = 0$ for all propagation constants of interest, that is

$$\text{fit}_{\text{tot}} = |\text{Det}[M(\beta_a^+)]| + |\text{Det}[M(\beta_b^+)]| + |\text{Det}[M(\beta_a^-)]| + |\text{Det}[M(\beta_b^-)]|. \quad (10)$$

The fitness function [Eq. (10)] gives the error for a group of values representing all propagation constants (forward and backward directions) and the guide separation $2S_3$. The goal is to obtain a group of propagation constants that makes Eq. (10) as close to zero as possible.

7 Nonlinear and Nonreciprocal Coupler Design

The FD-BPM formalism discussed in Sec. 4 is utilized here on the design and simulation of a directional coupler consisting of nonlinear and magneto-optic materials. The design of the coupler takes into account that material parameters (such as nonlinear coefficients, magneto-optic

constants, and refractive indices) cannot be varied at will. Therefore, only a set of commonly used values in the literature will be used here, and these values will not be optimized during the calculations. Guides A and B are both single-moded if taken in isolation. It is noteworthy that the refractive indices n_1 and n_5 are made slightly larger than n_3 in order to increase the field interaction with the nonreciprocal media. Since the coupler operation requires two different power levels, the design will also be divided in two separate sections. Only the parallel section of the structure is considered in the optimization procedure. The design algorithm is as follows.

7.1 Linear Operation

1. Coupling length (+z direction): defined as $L_{\pi(\text{linear})}^+$, guides A and B are phase matched.
2. Coupling length (−z direction): $L_{\pi(\text{linear})}^- = 0.5 \times L_{\pi(\text{linear})}^+$, guides A and B are not phase matched. This coupling length guarantees that even if power is partially transferred from guide B to guide A, it will be coupled back to guide B, exiting on port 2. This forward and backward L_{π} combination relaxes the optimization procedure and, in addition, maximizes the optical isolation in port 1.
3. The optimization is carried out with GA utilizing the analytical formalism for the multilayer structure. The optimized parameters are: guide separation $2S_3$, and guide thicknesses w_1 and w_2 . The coupling lengths $L_{\pi}^{\pm}$ are given by

$$L_{\pi(\text{linear})}^{\pm} = \frac{\pi}{|\beta_a^{\pm} - \beta_b^{\pm}|}. \quad (11)$$

Equation (11) is utilized by the GA, together with Eq. (10), as a fitness function subjected to the constraints defined in items 1 and 2 for the determination of the device length. The coupling length $L_{\pi(\text{linear})}^+$ will be used as the actual device length.

7.2 Nonlinear Operation

1. The nonlinear operation requires an extra care, and for this purpose the fitness function will instead be defined in terms of the power ratio P_A/P_B , as discussed later on in this section.
2. Recall that all geometrical parameters have already been defined during the optimization of the linear case. The only parameter left to be optimized is the nonlinear index increase, Δ . Moreover, note that the analytical formalism described in Sec. 5 does not account for nonlinear materials. In this case, the layer corresponding to the nonlinear material is considered as having the refractive index dependence $\bar{n}_2 = n_2 + \Delta$, where Δ is the refractive index increase due to the nonlinear effect. The parameter Δ can be optimized by the GA to produce the desired power ratio P_A/P_B . Once Δ is obtained, all that is left to be done is to find the corresponding optical power density that produces this particular refractive index increase.

3. The optimization of the nonlinear operation can be conveniently initiated in the $-z$ direction. The fitness function in this case is defined in terms of the combination of the symmetric and antisymmetric modes of the combined structure. Once both modes are found, they must be combined to produce the excitation field in guide A, i.e., $H_{y(\text{exc})} = H_y^{\text{sym}} + \text{sgn} \times H_y^{\text{antisym}}$, $\text{sgn}=1$ or -1 . This field is assumed at $z = L_\pi$ (see Fig. 1);
4. The excitation field must now be propagated towards the input end of the coupler, $H_{y(\text{in})}(z=0)$, which can be done as follows:

$$H_y^{\text{sym}} = \Re\{H_y^{\text{sym}} \exp(j\beta_1 L_z)\},$$

$$H_y^{\text{antisym}} = \Re\{\text{sgn} \times H_y^{\text{antisym}} \exp(j\beta_2 L_z)\},$$

$$H_{y(\text{in})} = H_y^{\text{sym}} + H_y^{\text{antisym}}.$$

The power density at $z=0$ is now obtained as follows

$$P_A = \frac{\beta}{\omega \epsilon_0} \int_0^\infty \xi_{xx} H_{y(\text{in})}^2 dx \quad \text{power in guide A,}$$

$$P_B = \frac{\beta}{\omega \epsilon_0} \int_{-\infty}^0 \xi_{xx} H_{y(\text{in})}^2 dx \quad \text{power in guide B,}$$

$$\xi_{xx} = \frac{\epsilon_{zz}}{\epsilon_{xx}\epsilon_{zz} - \delta^2}.$$

The fitness function in this case is defined as $\text{fit} = P_A/P_B$, where P_A and P_B are the optical power in guides A and B, respectively, at $z=0$. A similar procedure is adopted in the forward direction.

8 Device Simulation

The physical and geometrical parameters of the device are summarized in Table 1. In all simulations the laser operation condition is assumed as cw. According to the design algorithm described in Secs. 7.1 and 7.2, the first step consists in the design of the linear operation. Guides A and B are initially considered in isolation and the thickness w_1 is fixed at $1.2 \mu\text{m}$ (maximum thickness for single-mode operation). Thickness w_2 is calculated in such a way as to allow perfect phase matching condition between both guides, viz., $w_2 = 1.23 \mu\text{m}$. This is the only situation where guides A and B are considered in isolation, all other calculations assume the coupler as a compound structure. The next step is the optimization of the guide separation $2S_3$ and the device length L_π , using the formalism described in Sec. 5. These parameters must necessarily be obtained via the GA since their values must satisfy the operation requirements for both propagation directions. The GA optimized values for these parameters are $2S_3 = 0.75 \mu\text{m}$, $L_\pi^{+(\text{linear})} = 1374.151 \mu\text{m}$ and $L_\pi^{-(\text{linear})} = 687 \mu\text{m}$. At this point, all pertinent parameters for linear operation are already known. Therefore, the optimized structure can now

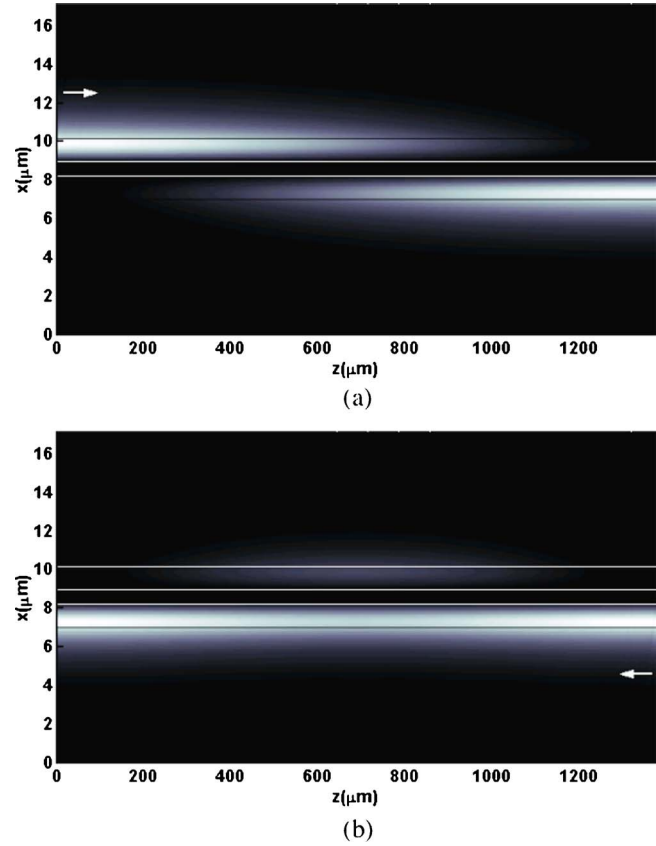


Fig. 5 Optical intensity distribution from the wide-angle FD-BPM simulation of the optimized structure under linear operation condition. (a) $+z$ direction, $L_\pi^{+(\text{linear})} = 1374.151 \mu\text{m}$ and (b) $-z$ direction, $L_\pi^{-(\text{linear})} = 687 \mu\text{m}$. Arrows indicate the propagation direction, parallel lines indicate dielectric interfaces.

be simulated with the FD-BPM formalism to verify its operation condition. The FD-BPM computational domain has dimensions $L_x = 17.18 \mu\text{m}$ and $L_z = 1375 \mu\text{m}$, and spatial discretization steps $\Delta x = 0.01 \mu\text{m}$ and $\Delta z = 0.2 \mu\text{m}$. A qualitative plot of the results is shown in Fig. 5. Observe that in Fig. 5(a) there is total transfer of power to port 4, with a coupling length $L_\pi^{+(\text{linear})} \approx 1375 \mu\text{m}$. In Fig. 5(b), on the other hand, the power is partially transferred from guide B to guide A, with the maximum occurring at $z = L_\pi^{+(\text{linear})}/2$ as required. The power density level for these simulations is 0.1 mW/mm . The isolation obtained for port 1 is -30.66 dB .

Next we proceed with the optimization of the nonlinear operation. This requires the minimization of the power ratio P_A/P_B for the backward propagation. The nonlinear index variation that best satisfies this prerequisite is $\Delta = 0.002$ which was found with the GA. The power density that gives an average index increase of this magnitude is 10.5 mW/mm . The propagating field in guides A and B are assumed phase mismatched in the $+z$ direction. The laser excitation is assumed as cw, therefore the backward propagating wave must experience the refractive index increase produced by the forward wave plus the index increase produced by the backward wave itself as a result of reflection from port 3. Therefore, the reflected power can affect con-

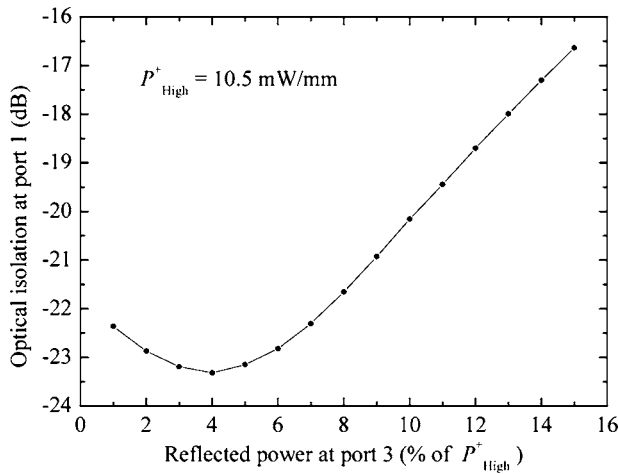


Fig. 6 Isolation of port 1 as a function of the reflected power at port 3.

siderably the isolation characteristics of the device and an optimum operation point needs to be found. The isolation dependence on the reflected power can be seen in Fig. 6. This figure clearly shows that the best-case scenario requires a 4% power reflection at port 3, resulting in a port 1 isolation of ≈ -23.32 dB. The BPM simulations for this case are shown in Fig. 7. All simulation parameters are kept

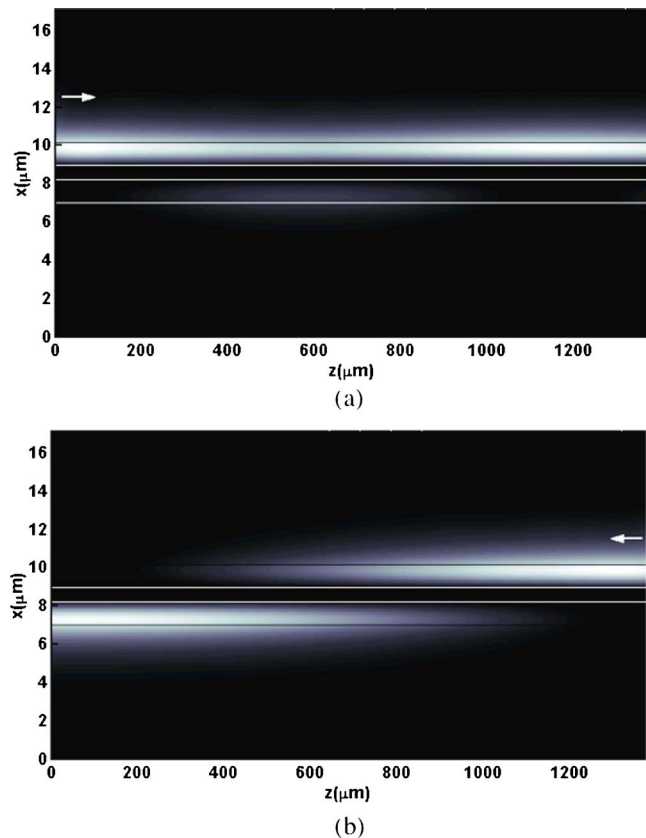


Fig. 7 Optical intensity distribution from the wide-angle FD-BPM simulation of the optimized structure under nonlinear operation condition. (a) +z direction; (b) -z direction. Arrows indicate the propagation direction, parallel lines indicate dielectric interfaces.

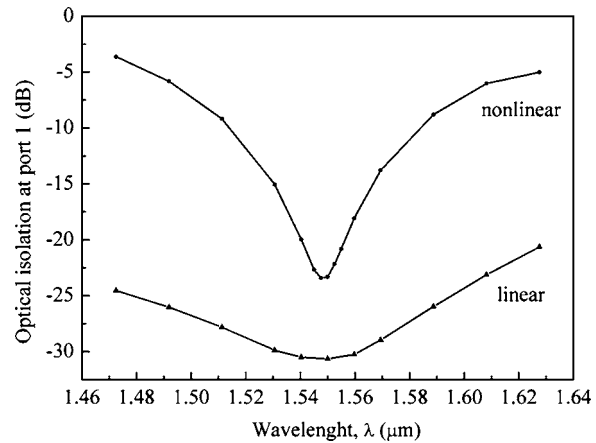


Fig. 8 Optical isolation of port 1 as a function of wavelength for the linear and nonlinear operation conditions.

the same as in the previous case. The coupling lengths obtained with the BPM simulation for the +z and -z directions are $\approx 583 \mu\text{m}$, and $\approx 1322 \mu\text{m}$, respectively, instead of $687 \mu\text{m}$ and $1375 \mu\text{m}$ as one would expect. This difference arises since only one degree of freedom was left available for the optimization procedure, namely the index variation Δ . We decided to keep the nonlinearity constant n_{2S} fixed since this is a material parameter that can not be found at will. Even though the coupling lengths do not totally satisfy the design criteria, the penalty imposed on the device performance as an optical isolator is not compromised.

Another important parameter to be investigated for the optimized structure is the isolation sensitivity of port 1 with respect to the wavelength. This dependence can be seen in Fig. 8 for linear and nonlinear operation conditions. As far as linear operation condition is concerned, the structure performs reasonably well for a large range of frequencies, which makes it a good candidate for optical isolator in wavelength division multiplexing (WDM) applications. The nonlinear operation condition, on the other hand, presents a much shorter isolation range. Considering a -20-dB optical isolation as a satisfactory value, this leaves an isolation window of about 20 nm, ranging from $1.54 \mu\text{m}$ to $1.56 \mu\text{m}$. Considering a typical channel separation of 0.8 nm present in current WDM systems, the device should be able to accommodate up to 25 channels. If a hybrid operation is desired, that is, a combined linear and nonlinear operation condition, the nonlinear case should be set as an upper bound for the device capacity.

9 Conclusions

The design of a new integrated optical device capable of operating simultaneously as an optical isolator and optical switch was discussed in this paper. The device performance was optimized with a GA and tested with a wide-angle FD-BPM. The optical isolation of the excitation port was -30.66 dB for linear operation, and -23.32 dB for nonlinear operation. Under linear operation conditions, the device proved to be reasonable insensitive to variations on the wavelength of operation. This is an especially attractive characteristic for WDM applications. Under nonlinear op-

eration, the performance is more significantly affected by variations on the wavelength. Using the latter case as upper bound for the hybrid linear/nonlinear operation, the device bandwidth becomes 20 nm, enough to accommodate up to 25 WDM channels.

Appendix

Nonzero elements for matrix $[M(\beta)]$ in Eq. (9) are

$$M_{11} = 1; \quad M_{12} = -\cos(\kappa_2 d_2/2); \quad M_{13} = -\sin(\kappa_2 d_2/2)$$

$$M_{21} = -j\xi_{zx}^{(1)}\beta + \gamma_1\xi_{zz}^{(1)};$$

$$M_{22} = j\xi_{zx}^{(2)}\beta \cos(\kappa_2 d_2/2) - \kappa_2 \xi_{zz}^{(2)} \sin(\kappa_2 d_2/2)$$

$$M_{23} = j\xi_{zx}^{(2)}\beta \sin(\kappa_2 d_2/2) + \kappa_2 \xi_{zz}^{(2)} \cos(\kappa_2 d_2/2)$$

$$M_{32} = \cos(\kappa_2 d_2/2); \quad M_{33} = -\sin(\kappa_2 d_2/2)$$

$$M_{34} = -\exp[-\gamma_3 S_3]; \quad M_{35} = -\exp[\gamma_3 S_3]$$

$$M_{42} = -j\xi_{zx}^{(2)}\beta \cos(\kappa_2 d_2/2) - \kappa_2 \xi_{zz}^{(2)} \sin(\kappa_2 d_2/2)$$

$$M_{43} = j\xi_{zx}^{(2)}\beta \sin(\kappa_2 d_2/2) - \kappa_2 \xi_{zz}^{(2)} \cos(\kappa_2 d_2/2)$$

$$M_{44} = (j\xi_{zx}^{(3)}\beta - \xi_{zz}^{(3)}\gamma_3)\exp[-\gamma_3 S_3]$$

$$M_{45} = (j\xi_{zx}^{(3)}\beta + \xi_{zz}^{(3)}\gamma_3)\exp[\gamma_3 S_3]$$

$$M_{54} = \exp[\gamma_3 S_3]; \quad M_{55} = \exp[-\gamma_3 S_3]$$

$$M_{56} = -\cos(\kappa_4 d_4/2); \quad M_{57} = -\sin(\kappa_4 d_4/2)$$

$$M_{64} = -(j\xi_{zx}^{(3)}\beta + \xi_{zz}^{(3)}\gamma_3)\exp[\gamma_3 S_3]$$

$$M_{65} = -(j\xi_{zx}^{(3)}\beta + \xi_{zz}^{(3)}\gamma_3)\exp[-\gamma_3 S_3]$$

$$M_{66} = j\xi_{zx}^{(4)}\beta \cos(\kappa_4 d_4/2) - \xi_{zz}^{(4)}\kappa_4 \sin(\kappa_4 d_4/2)$$

$$M_{67} = j\xi_{zx}^{(4)}\beta \sin(\kappa_4 d_4/2) + \xi_{zz}^{(4)}\kappa_4 \cos(\kappa_4 d_4/2)$$

$$M_{76} = \cos(\kappa_4 d_4/2); \quad M_{77} = -\sin(\kappa_4 d_4/2); \quad M_{78} = -1$$

$$M_{86} = -j\xi_{zx}^{(4)}\beta \cos(\kappa_4 d_4/2) - \xi_{zz}^{(4)}\kappa_4 \sin(\kappa_4 d_4/2)$$

$$M_{87} = j\xi_{zx}^{(4)}\beta \sin(\kappa_4 d_4/2) - \xi_{zz}^{(4)}\kappa_4 \cos(\kappa_4 d_4/2)$$

$$M_{88} = j\xi_{zx}^{(5)}\beta + \xi_{zz}^{(5)}\gamma_5.$$

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Licinius D. S. Alcantara was a data processing technologist at High Educational Center of Para (CESUPA) in 1995. He received his BS and MSc degrees from Federal University of Para (UFPA) in 1998 and 1999, respectively, and his PhD degree from University of São Paulo (Engineering High School of São Carlos, EESC/USP) in 2004, all in electrical engineering. Since September 2004, he has been with the Electrical and Computer Engineering Department (DEEC) at UFPA, conducting teaching and research activities. His main interests are numerical methods for electromagnetic wave propagation in complex media.



Marcos A. C. Lima received his BSc degree in electrical engineering from Federal University of Goiás, Goiânia, Brazil, in 1996, his MSc degree in electrical engineering from Federal University of Santa Catarina, Florianópolis, Brazil, in 1999, and his PhD degree in electrical engineering from School of Engineering of São Carlos, University of São Paulo, São Carlos, Brazil, in 2005. His areas of interest include evolutionary computation and its application for resource allocation problems in wireless and optical networks, design of micro-waves, and optical devices.

Amílcar C. César received his BSc degree in electrical engineering from University of São Paulo, São Carlos, Brazil, in 1976, and his MSc and PhD degrees in electrical engineering from State University of Campinas, Brazil, in 1982 and 1990, respectively. Since 1977, he has been with the Electrical Engineering Department, School of Engineering of São Carlos, University of São Paulo, where he is currently an associate professor. His areas of interest include computational modeling of waveguides for microwaves, millimeter waves, and optical applications.



Ben-Hur V. Borges received his BSc degree in electrical engineering from Universidade do Vale do Rio Doce, Gov. Valadares, Brazil, in 1987, his MSc degree in electrical engineering from University of São Paulo, São Carlos, Brazil, in 1992, and his PhD degree in electrical engineering from Drexel University, Philadelphia, Pennsylvania, in 1997. From 1997 to 1998 he was a postdoctoral researcher at the São Carlos School of Engineering of the University of São Paulo, in São Carlos, Brazil. Since 1998, he has been an assistant professor at the University of São Paulo in São Carlos. He is a member of the International Compumag Society. His research interests include the investigation of wave propagation in complex media including left-handed metamaterials, and modeling of devices for communication and sensing applications.



Fernando L. Teixeira received his BS and MS degree in electrical engineering from the Pontifical Catholic University of Rio de Janeiro (PUC-Rio), Brazil, in 1991 and 1995, respectively, and his PhD degree in electrical engineering from the University of Illinois at Urbana-Champaign in 1999. From 1999 to 2000, he was a postdoctoral research associate with the Research Laboratory of Electronics, Massachusetts Institute of Technology (MIT), Cambridge. Since 2000, he has been an assistant professor at the ElectroScience Laboratory and the Department of Electrical Engineering, The Ohio State University, Columbus. His research interests include analytical and numerical techniques for wave propagation and scattering problems in communication, sensing, and devices applications. Dr. Teixeira has edited one book entitled *Geometric Methods for Computational Electromagnetics* (PIER 32, EMW: Cambridge, MA, 2001), and has (co)authored over 60 journal articles and book chapters, and over 80 conference publications in those areas. He was awarded many prizes for his research, including the NSF CAREER Award in 2004 and the triennial USNC/URSI Booker Fellowship in 2005. He is a member of Phi Kappa Phi and Sigma Xi, a senior member of IEEE, and a full member of URSI Commission B.